

DUPlet: Dynamic User Modeling for Personalized Cover Letter Generation

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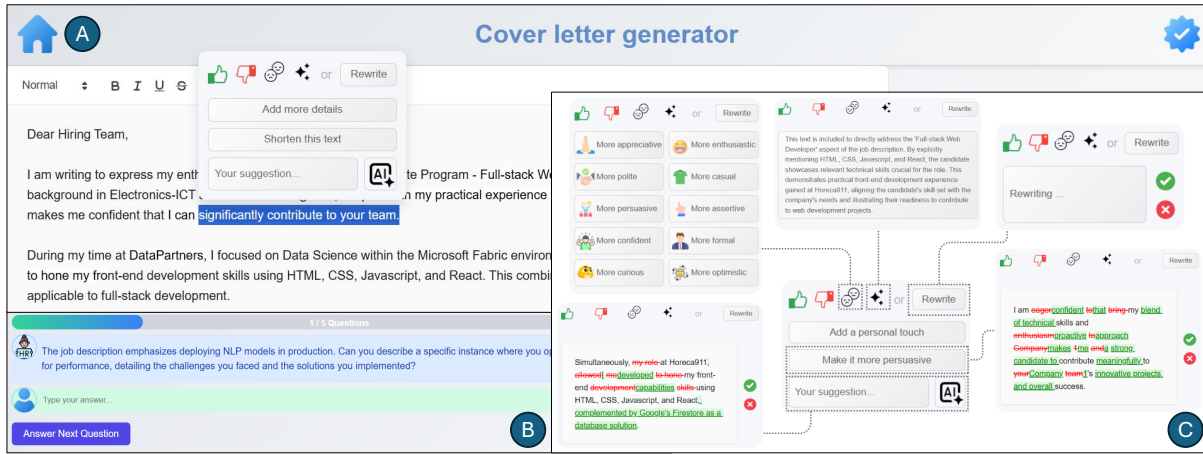


Figure 1: DUPlet is a novel interactive system that helps co-write cover letters with AI. (A) The user interface with AI-based features. (B) A Q&A module that contributes to an initial draft based on job-relevant questions. (C) DUPlet logs user interactions and preferences and builds a dynamic user model stored using natural language summaries for user-specific personalization.

Abstract

Composing tailored cover letters for specific job applications is challenging, particularly when applying for multiple jobs. We present *DUPlet*, an interactive system that leverages large language models (LLMs) to produce more personalized and authentic cover letters. The key novelty of our system lies in dynamic user modeling that leverages a natural-language description of the user's profile and preferences, which adapts over time based on provided input and interactions with the system. Based on a resume, job description, and a series of LLM-generated, job-specific questions, *DUPlet* creates an initial cover letter draft that users can then iteratively refine. We conducted a preliminary evaluation of *DUPlet* by simulating user interactions with LLMs and using the LLM-as-a-judge paradigm to

rate the cover letters on dimensions, such as personalization and authenticity. Results indicate that cover letter variants that involve the user and leverage a dynamically refined natural-language user profile tend to receive higher ratings.

CCS Concepts

• Human-centered computing → User models; • Computing methodologies → Natural language generation.

Keywords

User Modeling, Human-AI Interaction, Personalization, Large-Language Models, Cover Letter Generation



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1 Introduction

While Artificial Intelligence (AI) can revolutionize the job search process, over-reliance on AI raises concerns, such as a loss of authenticity for the applicants [8] and hallucinations [8, 23]. We present Dynamic User Modeling for Personalized Cover Letter Generation (*DUPLet*), an LLM-powered system that generates tailored cover letters from a resume, a job posting, and targeted questions. Unlike a static profile that holds a fixed user representation, *DUPLet* creates and uses a dynamic user model expressed in natural language that captures user preferences, intentions, and stylistic attributes, which provides additional interpretability. We conducted a preliminary evaluation using LLM-generated applicant personas [14] ($N=20$) from realistic resumes [5]. We then simulated a cover letter creation process across two domains (“Data Science” and “Finance”) in which each simulated user prepared three cover letters and evaluated them using the LLM-as-a-judge paradigm [21]. Preliminary results show improvements for personalization, authenticity, and truthfulness. Variants that included additional user input ranked higher on the 7-point Likert scale. While this shows potential, despite using validated methodologies [14, 21], our findings are limited by reliance on LLMs for generation and evaluation. Consequently, future work should validate these results with real-world users.

2 Related Work

Personalization involves adapting content or recommendations to a user’s specific needs [2]. Generally, LLM output can be adapted through (1) fine-tuning or (2) prompt engineering. For personalization, other alternatives include retrieval-augmented generation (RAG) with parameter-efficient fine-tuning [13], which shows improvements on the Language Model Personalization (LaMP) benchmark [12]. However, because such an approach may increase latency, in *DUPLet* we opted to augment the input prompt with a natural-language user profile, which has been shown to enhance performance across a wide range of tasks [10, 19]. LLMs can assist candidates in the job search process through better job recommendations [22], tailored resumes [23], or mock interviews [15]. Cover letters, which are the use case in this paper, were also considered in prior work [1]. A prior study found that cover letters generated by GPT-4 were statistically indistinguishable from those written by a human, based on impression, readability, criteria satisfaction, and degree of detail [4]. *DUPLet*’s novelty lies in personalization using a dynamic user profile that evolves over time through interactions with the system and is stored as a natural-language summary.

3 System Overview

DUPLet (Figure 2) is an LLM-based system implemented using Gemini 2.0 Flash that leverages prompting to explore dynamic user modeling and personalized cover letter generation. It is implemented as a web application in which new users first upload their resumes, which are then transformed into an initial natural-language user profile by the *Dynamic User Modeling* module. When a user applies for a job, they paste the job description into the system and proceed to the *Q&A* module, which generates five relevant questions based on the user’s profile and the job posting. Next, the *Personalization* module generates an initial draft of the cover letter, which the user refines through an interactive UI component. Finally, when the user

has finished, the *Dynamic User Modeling* module updates the profile based on all interactions with the system.

3.1 Dynamic User Modeling

This module creates a user profile dynamically, i.e., based on user interaction and preferences, as an unstructured natural language summary with an approximate length of 200-500 words. The user uploads a PDF resume to mitigate the cold-start problem, since the system has no information. A multimodal LLM (Gemini 2.0 Flash) extracts relevant information from the resume file, such as name, language, education, work experience, and skills, and converts them to JSON stored in Long-Term Memory (LTM). The LTM stores all previously collected information (e.g., interaction history and answers from the *Q&A* module of previous job postings). The JSON format is transformed into an initial natural-language user summary that includes both basic user information and inferred preferences, goals, and stylistic attributes. This summary is stored in Short-Term Memory (STM), which is the natural language user profile used for personalization during the current session. Once the user has finished their cover letter, all the information from the buffer, which contains interactions from the current session (e.g., *Q&A* question-answer pairs, likes and dislikes, rewrites, and applied suggestions), is persisted to LTM. The initial user profile summary is updated using Chain-of-Thought (CoT) prompting and stored in STM. Our approach is similar to that of Yang et al. [20], who used CoT prompting to reason about the user at intermediate steps. However, *DUPLet* performs the reasoning steps within a single prompt rather than using multiple intermediate prompts, which is important for real-world deployment.

3.2 System Interactions

DUPLet aims to create more authentic and trustworthy cover letters by capturing user information from the interactions with the system. It generates the first version of the cover letter (known as *draft cover letter*), using both the user profile and the question-answer pairs from the *Q&A module*. The user can further refine it using different types of interactions supported by the UI (Figure 1). E.g., selecting a text will provide a pop-up offering several suggestions and actions, such as liking, disliking, and explaining fragments of the text, as well as options to adjust the tone. Furthermore, it proactively offers two system-proposed suggestions (e.g., “Shorten this text”) out of six predefined suggestions using Thompson Sampling (TS) [16], along with text input. We model the click-through rate as a beta distribution with $\alpha = count_{clicked} + 1$ and $\beta = count_{shown_but_not_clicked} + 1$. *DUPLet* samples from each distribution and shows the two with the largest values. This balances exploration (new suggestions) and exploitation (showing previously accepted ones) to adapt the UI to each user.

3.3 Personalization

This module uses the natural-language summary and the job description to create a personalized cover letter. We adopted a prompt-based personalization strategy [13] to incorporate non-parametric user information. *DUPLet* generates cover letters following best practices in professional resume writing, as discussed by Zinjad et al. [23]. While our prompt design primarily follows that of Prompt

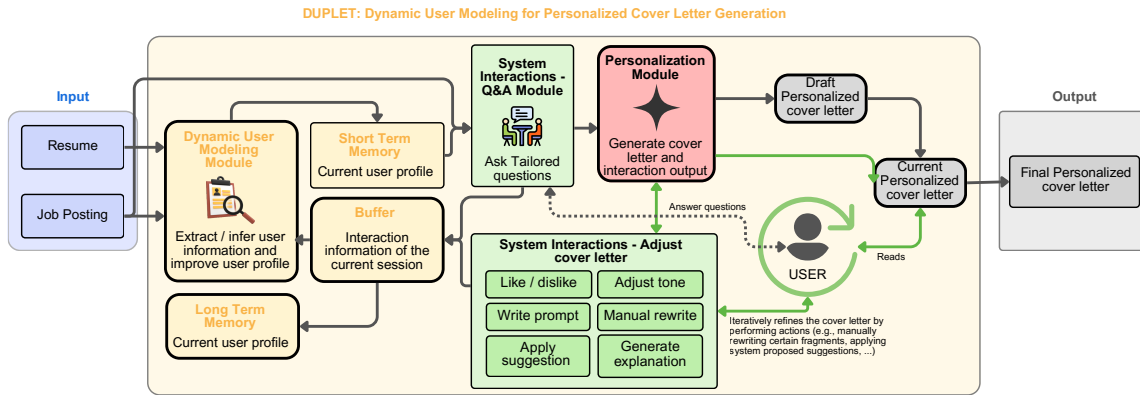


Figure 2: DUPLet: The system consists of three main components (Dynamic User Modeling, System Interactions, and Personalization) that work together to generate personalized cover letters and to refine the user profile over time.

Canvas [6], we changed the position of *Task* and *Context*, as providing the profile earlier benefits personalization [19].

4 Preliminary Evaluation

4.1 Study Design

We used an LLM (Gemini 2.0 Flash) to simulate realistic job applicant behavior and interact with the system. Each simulated test generates four cover letter variants: (1) **The baseline cover letter**, which is a cover letter tailored to the job without any personalization to an individual user. (2) **The static personalized cover letter**, which is a cover letter generated solely based on the natural language summary of the user, without any additional system interaction. (3) **The draft cover letter**, which is the cover letter generated after the user has completed the Q&A module. (4) **The final cover letter**, which is the cover letter generated through iterative refinement through the UI. Each variant was evaluated using an LLM-as-a-judge approach, producing ratings (7-point Likert scores) on five evaluation dimensions: personalization capabilities [18], quality of the generated content [18], job relevance [18], authenticity and style adaptation [11], and truthfulness [3]. The output also included a rationale (i.e., a concise explanation of the rating) and normalized word-frequency annotations (e.g., the number of words that do not match the user’s writing style or tone divided by the total number of words in the cover letter). We opted to use the LLM-as-a-judge approach for the preliminary evaluation, as prior work [18] shows it outperforms existing automated evaluation metrics for assessing personalization. To create realistic personas for the experiment, we used resumes from the *ResumeAtlas dataset* [5] and job postings from *LinkedIn Job Postings Dataset* [9]. The resumes are used both by the system for generation and the LLM-as-a-user to ground the simulated persona. The LLM was further instructed to generate two additional profiles: (1) A background profile that includes inferred attributes of the user’s career goals, intentions, strengths, weaknesses, work preferences, and hobbies; (2) A style profile, which describes the user’s communication and writing style, including tone, complexity, length, and sample sentences, based on the user’s resume and background profile. These two additional profiles are only visible to the LLM-as-a-user agent to generate

more consistent responses and not to the system generating the cover letters. The experiment includes 20 simulated personas and participants, 10 “Data Science” and 10 “Finance.”

4.2 Study Procedure

Each evaluation was initiated by providing a realistic resume and a job posting, which generated the initial natural-language user profile. Next, *DUPLet* generated the questions for the Q&A module, the baseline cover letter, and the static cover letter. At this point, the simulated participant was used to perform the following tasks using the resume profile, background profile, style profile, and all previous interaction data to ground their reasoning: (1) answer the five Q&A questions, (2) indicate three likes and dislikes on separate words or sentences in the *Draft cover letter*, (3) perform one rewrite action, (4) apply one system suggestion and reject or accept the system’s updated output, and (5) apply one self-made suggestion and reject or accept the system’s updated output. These interactions simulated a concise but realistic setting. Each simulated participant performs this process for three distinct job postings in their respective profession. All simulated participants in each subgroup (“Data Science” and “Finance”) follow the same process for the same three job postings in a fixed order. After the simulated participant completed this process, the LLM-as-a-judge, which takes the persona of an expert recruiter, evaluated the four cover letter variants based on the simulated participant’s resume, background, and style profile, as well as the job posting. The LLM-as-a-judge evaluated each cover letter and was instructed to use a 7-point Likert scale using a zero-shot CoT prompting strategy. It was also instructed to mark all words or sentences that do not align with the user’s writing style or tone, or those that are incorrect, unverifiable, or highly exaggerated. However, this analysis is beyond the scope of this paper and left for future work.

4.3 Results

We used the Friedman test to analyze repeated-measures data from 7-point Likert-scale ratings. Figure 3 shows the resulting ratings for three of the five dimensions. Results show a statistically significant difference ($p < 0.05$) between the cover letters on all dimensions:

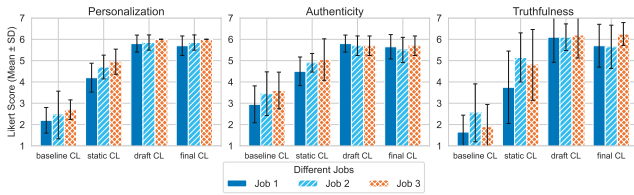


Figure 3: Mean and standard deviation of the 7-point Likert scores for personalization, authenticity, and truthfulness.

personalization ($p < 0.001$), quality ($p = 0.0016$), job relevance ($p = 0.048$), authenticity ($p < 0.001$), and truthfulness ($p < 0.001$). We further conducted post-hoc pairwise comparisons using the Wilcoxon signed-rank test [7, 17] (two-sided). For personalization, authenticity, and truthfulness, all pairwise comparisons, except for Draft vs Final, were significant. For Quality, no pairwise comparisons reached $p < 0.05$; for Job Relevance, the baseline differed from static, draft, and final, but other pairwise comparisons were not significant. We also investigated the impact of user profiles refined over time due to dynamic user modeling and hypothesized that the static cover letter for Job 3 would be rated higher than for Job 1. The Friedman test found a significant difference for personalization ($p = 0.0038$), authenticity ($p = 0.011$), and truthfulness ($p = 0.0024$). No significant effects were found for Quality and Job Relevance. Pairwise Wilcoxon signed-rank tests show Job 1 vs Job 3 differed significantly for personalization ($p = 0.0045$), authenticity ($p = 0.047$), and truthfulness ($p = 0.0088$). For truthfulness, Job 1 vs. Job 2 differed significantly ($p = 0.0026$). Other pairwise comparisons were not significant.

5 Discussion and Conclusion

DUPLet is an LLM-based system for generating personalized cover letters that combines user modeling and human-AI interaction. Our preliminary evaluation using LLM-generated personas and an LLM-as-a-judge paradigm indicates improvements in personalization, authenticity, and trustworthiness. While promising, we acknowledge the following limitations. Our implementation relied on Gemini 2.0 Flash for the system and most evaluations, except persona generation, which used ChatGPT (GPT-4). Using the same family of LLMs for generation and evaluation introduces circularity bias, potentially leading to self-preference and overestimating results. Furthermore, this also constrains the ecological validity of the study, and follow-up work should investigate deployments with real users. Lastly, exploring other failure modes, such as hallucinations or over-personalization, is reserved for future work. Despite such limitations, *DUPLet* paves the way for AI-assisted writing systems that better adapt to its users.

Acknowledgments

DUPLet's UI uses icons obtained under a premium license from Flaticon.com.

References

- [1] Tarannum Alimahmad Bloch and Ibrahim Inusa. 2025. GPT-3-Based AI Cover Letter Generator: A Feasibility Study & Implementation. In *Proc. of International Conference on Recent Trends in Machine Learning, IoT, Smart Cities and Applications*. Springer Nature Singapore, 1–13.
- [2] Agustí Cerrillo. 2025. The personalization of digital services: How is it being driven by European Union Member States? *Digital Government: Research and Practice* (May 2025). doi:10.1145/3734697
- [3] Jin Chen, Zheng Liu, Xu Huang, Chenwang Wu, Qi Liu, Gangwei Jiang, Yuanhao Pu, Yuxuan Lei, Xiaolong Chen, Xingmei Wang, Kai Zheng, Defu Lian, and En-hong Chen. 2024. When large language models meet personalization: perspectives of challenges and opportunities. <http://dx.doi.org/10.1007/s11280-024-01276-1>. *World Wide Web* 27, 4 (June 2024). doi:10.1007/s11280-024-01276-1
- [4] Can D. Deveci, Baker Jason J., Binyamin Sikander, and Jacob Rosenberg. 2023. A comparison of cover letters written by ChatGPT-4 or humans. <https://pubmed.ncbi.nlm.nih.gov/38018708/>. *Danish Medical Journal* 70, 12 (2023), A06230412.
- [5] Ahmed Heakl, Youssef Mohamed, Noran Mohamed, Aly Elsharkawy, and Ahmed Zaky. 2024. ResumeAtlas: Revisiting Resume Classification with Large-Scale Datasets and Large Language Models. arXiv:2406.18125 [cs.CL] <https://arxiv.org/abs/2406.18125>
- [6] Michael Hewing and Vincent Leinhos. 2024. The Prompt Canvas: A Literature-Based Practitioner Guide for Creating Effective Prompts in Large Language Models. arXiv:2412.05127 [cs.AI] <https://arxiv.org/abs/2412.05127>
- [7] Mathias Jesussek. 2025. Wilcoxon signed-rank test - DATAtab Tutorial. <https://datatab.net/tutorial/wilcoxon-test>. Accessed: 2025-07-20.
- [8] Justin Kaashoek, Manish Raghavan, and John J. Horton. 2024. The Impact of Generative AI on Labor Market Matching. <https://mit-genai.pubpub.org/pub/4t8pq0t6>. *An MIT Exploration of Generative AI* (mar 27 2024).
- [9] rajatray0502. 2023. LinkedIn Job Postings Dataset. <https://www.kaggle.com/datasets/rajatray0502/linkedin-job-2023>. Dataset licensed under CC BY-SA 4.0. Accessed: 2025-07-11.
- [10] Chris Richardson, Yao Zhang, Kellen Gillespie, Sudipta Kar, Arshdeep Singh, Zeynab Raeesy, Omar Zia Khan, and Abhinav Sethy. 2023. Integrating Summarization and Retrieval for Enhanced Personalization via Large Language Models. arXiv:2310.20081 [cs.CL] <https://arxiv.org/abs/2310.20081>
- [11] Alireza Salemi, Julian Killingback, and Hamed Zamani. 2025. ExPerT: Effective and Explainable Evaluation of Personalized Long-Form Text Generation. <https://arxiv.org/abs/2501.14956>. arXiv:2501.14956 [cs.CL]
- [12] Alireza Salemi, Sheshera Mysore, Michael Bendersky, and Hamed Zamani. 2024. LaMP: When Large Language Models Meet Personalization. arXiv:2304.11406 [cs.CL] <https://arxiv.org/abs/2304.11406>
- [13] Alireza Salemi and Hamed Zamani. 2025. Comparing Retrieval-Augmentation and Parameter-Efficient Fine-Tuning for Privacy-Preserving Personalization of Large Language Models. <https://arxiv.org/abs/2409.09510>. arXiv:2409.09510 [cs.CL]
- [14] Andreas Schuller, Doris Janssen, Julian Blumenröther, Theresa Maria Probst, Michael Schmidt, and Chandan Kumar. 2024. Generating personas using LLMs and assessing their viability. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. doi:10.1145/3613905.3650860
- [15] Hongda Sun, Hongzhan Lin, Haiyu Yan, Yang Song, Xin Gao, and Rui Yan. 2025. MockLLM: A Multi-Agent Behavior Collaboration Framework for Online Job Seeking and Recruiting. <https://arxiv.org/abs/2405.18113>. arXiv:2405.18113 [cs.CL]
- [16] William R Thompson. 1933. On the likelihood that one unknown probability exceeds another in view of the evidence of two samples. <https://doi.org/10.1093/biomet/25.3-4.285>. *Biometrika* 25, 3/4 (1933), 285–294.
- [17] Univ. of Maryland School of Nursing. 2019. Mann-Whitney U and Wilcoxon Signed-Ranks Tests. <https://cf.son.umaryland.edu/NRSRG795/V2/module5/subtopic2.htm>.
- [18] Yaqing Wang, Jiepu Jiang, Mingyang Zhang, Cheng Li, Yi Liang, Qiaozhu Mei, and Michael Bendersky. 2023. Automated Evaluation of Personalized Text Generation using Large Language Models. <https://arxiv.org/abs/2310.11593>. arXiv:2310.11593 [cs.CL] <https://arxiv.org/abs/2310.11593>
- [19] Bo Wu, Zhiwei Shi, Hamid Aghakhani Rahmani, Vighnesh Ramineni, and Emine Yilmaz. 2024. Understanding the Role of User Profile in the Personalization of Large Language Models. arXiv (2024). <https://arxiv.org/abs/2406.17803>
- [20] Fan Yang, Yong Yue, Gangmin Li, Terry R Payne, and Ka Lok Man. 2024. Chain-of-thought prompting empowered generative user modeling for personalized recommendation. *Neural Computing and Applications* 36, 34 (2024), 21723–21742. doi:10.1007/s00521-024-10364-2
- [21] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in Neural Information Processing Systems* 36 (2023), 46595–46623. <https://arxiv.org/abs/2306.05685>
- [22] Zhi Zheng, Zhaopeng Qiu, Xiao Hu, Likang Wu, Hengshu Zhu, and Hui Xiong. 2023. Generative Job Recommendations with Large Language Model. arXiv:2307.02157 [cs.IR] <https://arxiv.org/abs/2307.02157>
- [23] Saurabh Bhausaheb Zinjad, Amrita Bhattacharjee, Amey Bhilegaonkar, and Huan Liu. 2024. ResumeFlow: An LLM-facilitated Pipeline for Personalized Resume Generation and Refinement. In *Proc. of ACM SIGIR Conference on Research and Development in Information Retrieval*. <https://arxiv.org/abs/2402.06221>